



Cyethack Solutions
"Your Privacy, Our Priority"

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DJANGO DEVELOPER ASSIGNMENT

Overview

Cyethack Solutions is building next-generation **AI-powered surveillance and threat detection platforms** for defence, government, and enterprise clients.

This assignment evaluates your ability to:

- Design and build **scalable Django applications**
 - Develop **secure REST APIs**
 - Work with **NoSql/SQL databases**
 - Apply **basic security and networking concepts**
 - Write clean, maintainable backend code
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Objective

Build a **Backend API System** for a simplified **Threat Monitoring & Alert Management Platform** using **Django + Django REST Framework (DRF)**.

The system should allow:

- User authentication
 - Threat/event ingestion
 - Alert generation
 - Secure data access via APIs
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Problem Statement

You are tasked with building a backend service that ingests **security events** from multiple sources and exposes APIs for managing and viewing **alerts**.

Think of this as a **backend module** that could later integrate with:

- Surveillance systems
 - SIEM tools
 - AI-based threat detectors
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Functional Requirements

1. Authentication & Users

- Implement **user authentication**
 - Use Django authentication or JWT-based auth
 - User roles:
 - Admin
 - Analyst
 - Role-based access:
 - Admin → full access
 - Analyst → read-only access to alerts
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2. Threat/Event Management

Create APIs to:

- Ingest security events (POST)
- Store events in the database

Each event should include:

- Source name (string)
- Event type (e.g., intrusion, malware, anomaly)
- Severity (Low / Medium / High / Critical)
- Description
- Timestamp (auto-generated)



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3. Alert System

- Automatically generate an **alert** when:
 - Severity = High or Critical
- Each alert should contain:
 - Reference to the event
 - Alert status (Open / Acknowledged / Resolved)
 - Created timestamp

APIs required:

- List alerts
 - Filter alerts by severity and status
 - Update alert status (Admin only)
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4. Database Design

- Use **PostgreSQL or SQLite or NoSql (MongoDB)**
 - Proper schema design
 - Optimized queries (avoid N+1 problems)
 - Use indexes where appropriate
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5. REST API Requirements

- Follow REST best practices
 - Proper HTTP status codes
 - Input validation and error handling
 - Pagination for list APIs
 - Clean and consistent API responses
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Technical Constraints

- Backend: **Python + Django**
 - APIs: **Django REST Framework**
 - Database: SQL-based / MongoDB
 - Version control: **Git**
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Security & Best Practices (Important)

You will be evaluated on:

- Secure handling of authentication tokens
 - Input validation to prevent SQL injection & misuse
 - Avoiding sensitive data exposure
 - Proper permission checks
 - Clean separation of concerns (views, serializers, models)
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Optional Enhancements (Bonus Points ★)

- JWT authentication
 - Basic rate limiting
 - Logging of critical actions
 - API documentation (Swagger / Postman)
 - Docker setup
 - Basic unit tests
 - Simple deployment steps (AWS / EC2 / Render / Railway)
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Deliverables

Mandatory

- GitHub repository link
- Complete Django project
- README.md containing:
 - Project overview
 - Setup instructions
 - API endpoints list
 - Assumptions made

Optional

- Postman collection
- Architecture diagram
- Sample `.env.example` file

Evaluation Criteria

Area	Weight
Django & DRF Fundamentals	★★★★
Database Design & Queries	★★★★
API Design & Clean Code	★★★★
Security Awareness	★★★★
Git & Project Structure	★★★



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Time Expectation

- **3 Days**
 - Focus on **quality, clarity, and correctness**
 - Submission before **25 DEC 2025 EOD**
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What We Are Looking For

- Strong Django & REST API fundamentals
 - Ability to think from a **security-first mindset**
 - Clean, readable, production-oriented code
 - Practical understanding of backend systems
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Submission

- Share the **GitHub repository link**
- Ensure the project runs locally without issues

 **Apply Here:**

<https://forms.gle/3MAPfJQot4dX5jWy6>